claude-windows / 45ea3a91-654d-4972-9cd3-65f35db54caf

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\subagents\workflows\wf_2a64bc88-2a8\agent-a9d1716e78cf84935.jsonl`
- SHA-256: `1913f7fd28a63b3b29fffd50e0e24b2a156333b1e53f2dfa49882e79def3e106`
- Source modified: `2026-06-10T06:11:16+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_2a64bc88-2a8`
- session_id: `45ea3a91-654d-4972-9cd3-65f35db54caf`

Transcript

1. user (2026-06-10T06:09:07.937Z)

```
CONTEXT ? two sibling repos on a Windows machine:
- INDEXER: C:/Users/avidu/Projects/badminton-highlight-indexer ? local AI badminton(->racket-sports) highlight indexer + rally detector. FastAPI + SQLite + ffmpeg + GPU CV via WSL; Gemini = paid cloud AI. docs/ tree is rich. Current branch docs/next-steps-multisport (master = main branch).
- COLLECTOR: C:/Users/avidu/Projects/sports-data-collector ? golden-label acquisition/curation/ingestion CLI (system-of-record candidate for training corpus).
Project state (June 2026): scaffolding complete; owned-model roadmap agreed (docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md) but NO platform/owned-model implementation started; next committed platform slice = async job model. Licensing guardrail: MIT/Apache-2.0/BSD only for code+data+weights; paid LLMs are inference-only, never training-data sources.
ENVIRONMENT RULES: this dev environment has NO GPU/torch/cv2 ? do NOT attempt to run the video pipeline or import torch/cv2 modules. Test suites are offline/fully-mocked and SAFE to run. You are READ-ONLY: never edit/create/delete repo files; running tests and git read commands is allowed.
QUALITY BAR: cite file paths (and line numbers where possible) for every finding. Report what IS in the code/docs, not what you assume. Be skeptical and concrete. Your final structured output is consumed programmatically by an orchestrator ? return raw data, not prose for a human.
ROLE: adversarial verifier. An auditor ("docs-strategy") claimed the finding below. Try to REFUTE it against the actual files. Default to isReal=false if the evidence does not hold up, the behavior is documented/deliberate (e.g. listed in docs/CODE_MAP.md gotchas or DEFERRED_HARDENING_PLAN.md as a known deferral ? unless the auditor adds genuinely NEW consequence), or the severity is materially overstated.
FINDING: {"title":"'The next slice' is answered three different ways: async job model vs Phase-0 M0.x vs packaging-first","severity":"high","area":"roadmap sequencing","file":"C:/Users/avidu/Projects/badminton-highlight-indexer/CLAUDE.md:20","evidence":"CLAUDE.md:20: 'Next slice = async job model (DEFERRED_HARDENING_PLAN.md #16), single-tenant' (also checkpoint README:30: 'First implementation slice = async job model'). docs/NEXT_STEPS.md:28 puts '[owner greenlight] Phase-0 (M0.x), $0/local' (M0.4 harness, M0.1 corpus, M0.3 compliance) as prioritized step 1, relegating async to a footnote (line 46: 'Committed next PLATFORM slice = async job model'). docs/DEFERRED_HARDENING_PLAN.md:12-14 recommends a different order entirely: 'Recommended sequence: #7 Packaging (lowest risk...) ? #13 Train/eval guardrails ? #16 Async jobs (largest)'. docs/COMMERCIALIZATION.md agrees with packaging-first (C9 packaging = P0 ?, C12 SaaS/async = P1 ?) and its §6 priority list ranks 'SaaS / multi-user / async foundations (C12)' 7th of 8 ('lower until monetization is firm'). The plan's footer (OWNED_MODEL_IMPLEMENTATION_PLAN.md:164-165) says 'the committed next platform slice remains the async job model...; only the M0.1?M0.4 foundation is safe to start.'","why_it_matters":"All docs agree the owner greenlights, but they disagree on what the committed first implementation item IS. A cold-start agent reading CLAUDE.md preps the async-jobs slice; one reading NEXT_STEPS preps M0.4; one reading DEFERRED_HARDENING/COMMERCIALIZATION preps packaging. The relationship between the 'committed platform slice' (async) and the 'recommended Phase-0' (M0.x) ? which runs first, or whether they're parallel tracks ? is stated nowhere."}
```

Steps: open the cited file(s) and any counter-evidence files; verify quotes/line claims; judge severity honestly. Cheap greps allowed; do NOT run the pipeline. Return isReal, confidence, and a 1-3 sentence verdict (with corrected detail if partially right).

2. user (2026-06-10T06:09:14.495Z)

```
# CLAUDE.md ? project memory / agent handoff

Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
repos: `sports-data-collector`, a VLC-based `rally-annotator`.

## Read these first
- **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
  contracts (the cold-start map; read before touching code).
- **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`, `docs/PMF_MARKET_RESEARCH.md`,
  `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also `docs/ROADMAP.md`
  (offline-segmenter narrative) and `docs/BACKLOG.md`.
- **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
  **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). Archive policy: only move
  **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level.

## Current state (June 2026)
- **Scaffolding complete**; the **post-scaffolding strategy foundation** is laid and merged
  (#52/#53/#55/#57/#58). **No platform/owned-model implementation has started.**
- **Next slice = async job model (`DEFERRED_HARDENING_PLAN.md` #16), single-tenant.**
  **Do not start implementation without explicit owner approval.**

## Architecture in one breath
- **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers / validators
  / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:*
  `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
- `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one helper).
- Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA
  user_version` migration ladder in `backend/infrastructure/database.py`.

## Committed guardrails (non-negotiable)
- **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a
  training data source** (terms/copyright). Training teachers/bases = **open Apache-2.0 only**.
- **Owned-model base:** Apache-2.0, video-capable (Qwen-VL / Gemma 4); **reject Gemma 3**
  (viral license). **Exclude minors** from the training pool; per-user training data needs a
  separate explicit opt-in.
- **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly future**
  (GDPR Art. 9).
- **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.

## Workflow conventions
- **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless asked.
- Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.8+; `ffmpeg` required.
- Never push to the default branch directly; descriptive commit messages.
- ?? This dev container lacks GPU / torch / cv2 / numpy ? the **full pipeline can't run here**.
  Verify pure-logic paths and hand machine-side checks (real video, GPU/WSL suite) to the owner.
```

3. user (2026-06-10T06:09:15.386Z)

```
# Next steps (live) ? owned-model + multisport

**Updated:** 2026-06-09 (end of the quality-iteration + multisport-recon session). **Authoritative roadmap:**
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`. **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`.
This file = the prioritized "what to do next," refreshed each session.

## The recommendation (one paragraph)
Build the **quality-first extraction harness** before anything else: a learned **TAL head** (ActionFormer/ASFormer, MIT)
over trajectory features is the robust path ? it **decouples from the multi-court "background games" confound** that the
unsupervised heuristic (and audio, and spatial court-gating) can't escape (all three tested + failed; it's a *temporal*
rally-vs-non-rally problem). **Framework = ms-swift, base = Qwen3-VL** (Gemma-4 AMBER/bench) come **later** for the
conversational/grounding upgrade ? start with the head. **On-device is deferred** (export-clean ops now so it stays
reachable). Everything stays **MIT/Apache-2.0/BSD** (code + data + weights). Bottleneck = **data + per-sport trajectory
detectors**, not model architecture.

## Latest findings (2026-06-09)
- **6-match badminton LOVO:** learned prototype **0.583** vs heuristic **0.480** (learned leads since n=3, AUC 0.83?0.92,
  decoupled from multi-court). The "contrast" metric (in-rally÷dead-time visibility) predicts heuristic F1 monotonically.
- **?? Multisport (Roora pickleball + TT):** labels **ingest cleanly** (`curate convert-cvat` ? 6 rallies/sport, sane).
  **WASB-transfer SURPRISE (directional, n=6 each ? TINY):** the badminton shuttle detector segments **TABLE TENNIS at
  F1 0.909** out-of-the-box (!) but **pickleball only 0.308**. ? **The per-sport-detector blocker is SPORT-DEPENDENT:** TT may
  be servable with WASB short-term; pickleball needs a proper (MIT/Apache) ball detector. Confirm/deny with more videos.
- **Owner field notes (2026-06-09):** camera height/distance **varied on purpose** (great generalization data); **multi-court
  adjacent-court SOUND dominates the signal** ? empirically reconfirms audio-foreground isolation won't work + the
  robustness-first learned-model path. Multi-court is the realistic (academy) environment, so this *is* the target distribution.

## Prioritized next steps
1. **[owner greenlight] Phase-0 (M0.x), $0/local:**
   - **M0.4 ? Gen-0 TAL harness** (first-class): productionize `rally_seq_proto.py` ? one-command train?eval?**ship-gate** over
     ActionFormer/ASFormer, reusing `metrics.match_intervals`. **Define the ship-gate target** (explicit F1@tIoU; floor = beat
     heuristic LOVO 0.480) ? "very high P/R" must be operationalized, not just "beat the floor."
   - **M0.1 ? corpus spine + the per-video event-index contract** (the conversational seam: extract ? event store ? text-to-SQL).
   - **M0.3 ? compliance:** purge Gemini-derived training labels (`training.label_candidates`); WASB **C3** own-weights/clean-swap (gates ship).
2. **Multisport thread:**
   - Ingest the Roora pickleball/TT rally CSVs into the corpus (collector `import-local`).
   - **Get more pickleball + TT videos** ? re-run the WASB-transfer eval to confirm/deny TT?0.91, pickleball?0.31 (n=6 is too few).
   - Once each sport has ?3 matches, run a **merged prototype LOVO across badminton + TT
```

(+ pickleball)** ? the "how would a
merged model do" experiment. Identify a clean MIT/Apache ball detector for sports where WASB doesn't transfer (pickleball).
3. **Housekeeping:** **rename repo off "badminton"** (multisport); confirm collector = corpus/event-store system-of-record.

Pointers / artifacts
- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md`
  (2026-06-09 banner) · Quality: `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Eval corpus: `output/human_lovo_manifest.json` (badminton n=6). Roora CSVs: `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv`;
  WASB-transfer trajectories: `%TEMP%\{pickleball,tabletennis}_traj.csv`. Code: `backend/eval/{eval_partial_labels,rally_seq_proto,calibrate_local}.py`.
- ? No owned-model implementation until owner greenlights a milestone. Committed next PLATFORM slice = async job model (single-tenant).

4. user (2026-06-10T06:09:16.172Z)

Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging

Status: ? Design for review ? **no code yet**. Owner-requested follow-up to the "config & identity consistency" tech-debt pass. Each item below becomes its own PR only after this doc is approved.

Why these three: they were cataloged as out-of-scope during the consistency pass because each is larger/design-heavy or needs the owner's machine. They are the remaining high-value hygiene items not already done (config/structure/Result/logging/reports) or in flight (detector abstraction, PR #49).

Recommended sequence: #7 Packaging (lowest risk, unblocks clean installs/CI) ? #13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs (largest, product-facing).

---

1. Packaging modernization (debt #7)

Problem
- No `pyproject.toml` on `master`. (PR #49's description lists packaging as "done", but it isn't present ? reconcile with the owner.)
- The root `setup.py` is a **diagnostics/dependency-checker**, not a PEP 517 build script ? its name collides with setuptools semantics and will confuse `pip`/`build`.
- Deps live only in `requirements.txt`; `obsreport` is a commented soft dep (BACKLOG) and `supervision<0.30.0` is pinned (debt #14).

Approach
- Add `pyproject.toml` (PEP 621): project metadata + runtime deps mirroring `requirements.txt`, with optional-dependency groups:
  - `[project.optional-dependencies] dev` ? `pytest`, coverage tooling.
  - `reporting` ? `obsreport` pinned to its git tag (`v0.1.3`, per BACKLOG).
  - `gpu` ? torch/torchvision (note: the cu124 `--extra-index-url` is **not** expressible in standard PEP 621 deps; document the install line or use a `[tool.uv]`/pip-conf approach rather than faking it).
- Console entry points: `indexer = backend.main:main` (so `indexer --server` works), optionally eval drivers.
- **Rename the diagnostic `setup.py`** (e.g. `scripts/doctor.py`) so it no longer shadows the build system; update README/`run_all_tests.py`/CI references.
- Choose a build backend (hatchling recommended) and confirm the `backend/` import root stays valid (keep current layout; no src/ move required).

Risks
- `setup.py` rename touches docs and any muscle-memory invocations ? grep first.

```
45      - Torch CUDA wheels can't be pinned cleanly in PEP 621; must document the
46        `--extra-index-url` install path (the current `setup.py` logic can move into docs).
47      - CI must switch to `pip install -e .[dev]`.
48
49      ### Verification
50      - `pip install -e .[dev]` in a clean venv ? `run_all_tests.py` green.
51      - `indexer --help` resolves via the entry point.
52
53      ---
54
55      ## 2. Train/eval separation guardrails (debt #13)
56
57      ### Problem
58      Separation is **by convention only**. `backend/eval/calibrate_local.py` and
59      `calibrate_wasb.py` do honest leave-one-video-out (LOVO), but nothing *enforces*
60      that a video used to train the learned segmenter
```

5. user (2026-06-10T06:09:22.743Z)

```
150     | 1 | Greenlight Phase-0 milestone(s) ? rec: **M0.1 corpus+event-index**, then **M0.4
harness** | awaiting owner |
151     | 2 | WASB C3 ? first-class action item (own-weights / clean swap) | **owner: yes** ? |
152     | 3 | Ratify licensing **MIT / Apache-2.0 / BSD** (code+data+weights) | **owner: yes** ?
|
153     | 4 | Collector = corpus + event-store system-of-record | **owner: confirm?** |
154     | 5 | Rename repo off "badminton" | **owner action** ? |
155     | 6 | Conversational-QA: build seam now, punt model | **rec accepted?** |
156
157     ## Open risks / unknowns (none block Phase-0 research)
158     Budget/cost to reach Gen-2 **unquantified** (largest risk) · ship-gate "very high P/R"
target undefined · per-sport clean
159     trajectory detectors not yet identified · base-model pretraining provenance unauditable
(risk-acceptance) · real VLM
160     VRAM/token cost · single-model-multisport-temporal generalization unproven · DPA + no-
train (minors-excluded) for cloud
161     training · on-device 4B latency on 8 GB unmeasured.
162
163     ---
164     *Design-only and reversible. Nothing starts until you greenlight a milestone. Sequencing
discipline holds: the committed
165     next platform slice remains the async job model (single-tenant); only the M0.1?M0.4
foundation is safe to start.*
166
```

6. user (2026-06-10T06:09:23.986Z)

```
8:golden-set vendoring spend, packaging/distribution, and the product-market-fit
36:| C9 | **Packaging / distribution** | **P0** | ? | No `pyproject.toml`; the diagnostic
`setup.py` shadows the build system. | Design in `DEFERRED_HARDENING_PLAN.md` (#7) ? **awaiting
your approval** before code. |
37:| C12 | **SaaS foundations (async/auth/tenancy)** | **P1** | ? | Sync `/api/process` blocks
on long videos; no auth/tenancy. | Multi-user/platform strategy in `PLATFORM_ARCHITECTURE.md`
(pluggable storage/compute, tenancy, orchestration); async slice = `DEFERRED_HARDENING_PLAN.md`
(#16). |
179:7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
180:8. **Polish & distribution** (C9): installer story, first-run path, codec/output, economics.
209:  the audit's AV1 recommendation is not yet implemented); C9 packaging tied to
221:- `DEFERRED_HARDENING_PLAN.md` ? packaging (C9), async (C12) designs (pending approval).
```

7. user (2026-06-10T06:09:33.437Z)

```
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
```

8. user (2026-06-10T06:09:33.982Z)

Carried from the readiness review; commercialization items above are tracked in parallel.

1. **Golden-set expansion + data-flywheel UX** (highest with new labeled videos):
   ingest the new batch, expand calibration/CV/regression baselines, make label
   contribution + safe retraining easy and non-contaminating (eval/training separation).
2. **Hybrid unification + observability hooks** (deferred item 7, now higher): extract
   shared logic from `tracknet_hybrid`/`wasb_hybrid` before heavy AI timing/metrics land.
3. **Detector/windowing quality iteration** with new data: re-tune velocity/merge/
   min-duration + `rally_gate`; validate WASB on real video; trial audio fusion (ADR-007).
4. **Deeper value layer** (beyond minimal exports, C10): per-rally tagging/notes ?
   `candidate_segments`, per-player/match stats, clip gallery, human-correction loop.
5. **Input-quality enforcement & guidance polish** (C11): more guideline dimensions ?
   validators/report rules; "your video is good enough" UX.
6. **Multi-sport maturation + recording best practices.**
7. **SaaS / multi-user / async foundations** (C12) ? lower until monetization is firm.
8. **Polish & distribution** (C9): installer story, first-run path, codec/output, economics.

Recommendation: with the mid-week labeled-video batch, spend ~80% on #1 + #3
(flywheel + quality iteration on real data), then unification (#2) and deeper value (#4).

---

7. Open questions needing owner / counsel sign-off ??

1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves
   insufficient (fallback). Quote-only.
2. **Per-checkpoint / dataset licenses** for WASB / TrackNetV3 / any VLM ? train-own or
   confirm each (C3).
3. **Exact Via LA (AVC) / Access Advance (HEVC) rates & free thresholds** ? confirm live
   figures with counsel before relying on the "small SaaS under threshold" reading (C5/C6).

9. user (2026-06-10T06:09:39.137Z)

Checkpoint: 2026-06 Strategy Foundation (post-scaffolding)

Date: June 2026
STATUS: COMPLETED
Trigger: Owner merged the post-scaffolding strategy docs ? #52 (config/identity cleanup), #53 (deferred-hardening design), #55 (commercialization tracker), #57 (PMF market study), #58 (platform architecture).

Purpose
Captures the state once the **strategic / direction-setting foundation** for the
post-scaffolding phase was laid: the commercial dimension, market/PMF, the multi-user
platform architecture (pluggable storage/compute + the owned-model layer), and the
deferred-hardening designs. **Strategy only ? no implementation started.** Provides a
regression-detection baseline for priorities/decisions as quality + implementation work begins.

Key living documents at checkpoint time
- **COMMERCIALIZATION.md** ? commercial tracker (dashboard C1?C14, grouped Pending ? In-flight ? Completed with a P0/P1/P2 column).
- **PMF_MARKET_RESEARCH.md** ? cited market study; beachhead = **India badminton academies**.
- **PLATFORM_ARCHITECTURE.md** ? multi-user "assistant coach" platform: pluggable `StorageBackend`/`ComputeBackend` (BYO), generic schema (coach overlay optional), owned video-QA model layer (§5A), training-backend survey + nanochat-style harness, committed legal guardrail, per-user-adapter direction, v1-vs-vision scoping.
- **DEFERRED_HARDENING_PLAN.md** ? packaging, train/eval guardrails, async job model

```
(#16) ? *pending approval*.
19      - **CODE_MAP.md / BACKLOG.md / ROADMAP.md** ? updated for the config/identity cleanup.
20
21      See git history around #52/#53/#55/#57/#58 for exact snapshots.
22
23      ## Committed decisions / non-negotiables recorded here
24      - **Paid frontier models (Gemini/OpenAI/Anthropic): inference / serving / fallback only
? never a training data source** (competing-model terms). Teachers/bases = open Apache-2.0 only.
25      - **Owned-model base = Apache-2.0-clean, video-capable** (Qwen2.5-VL-7B/32B, Qwen3-VL,
Gemma 4); **reject Gemma 3** (viral license clause); **minors excluded** from the training pool;
per-user training needs a separate opt-in.
26      - **v1 = rally detection + highlight compilation + per-user history + coach/user
correction loop** ("assistant-coach lite"); **gait/biomechanics strictly future** (Art. 9).
27      - **Generic data schema** (User ? VideoCollection ? Video ? Highlight ?
HighlightMetadata); coach ? athlete is an **optional overlay** (a market-fit hypothesis).
28
29      ## What comes next (per owner)
30      - First implementation slice = **async job model (DEFERRED #16)**, single-tenant ?
**pending explicit go-ahead** (no implementation started without approval).
31      - Quality iterations with new labeled videos; golden-set expansion + the **golden QA
eval set** (the prerequisite for the owned-model training harness).
32
33      When the first implementation slice or a quality milestone completes, add a new dated
34      checkpoint here (or under `quality-iterations/`).
35
```

10. user (2026-06-10T06:10:31.468Z)

```
1:# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging
3:**Status:** ? Design for review ? **no code yet**. Owner-requested follow-up to the
8:pass because each is larger/design-heavy or needs the owner's machine. They are the
13:#13 Train/eval guardrails (correctness of the data flywheel) ? #16 Async jobs
22:  but it isn't present ? reconcile with the owner.)
90:## 3. Async job model (debt #16)
96:### Approach (minimal first slice ? single self-hosted box)
123:- **Unify hybrid segmenters (#7 in LOW_REGRET):** owner-deferred.
125:  need a GPU/WSL box and real video ? owner-machine work, tracked separately.
```

11. user (2026-06-10T06:10:32.588Z)

```
docs\archives\COMMERCIAL_READINESS_REVIEW.md:90:   - Async job model, progress, storage,
auth/tenancy basics.
docs\DEFERRED_HARDENING_PLAN.md:1:# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs,
Packaging
docs\DEFERRED_HARDENING_PLAN.md:13:#13 Train/eval guardrails (correctness of the data flywheel)
? #16 Async jobs
docs\DEFERRED_HARDENING_PLAN.md:90:## 3. Async job model (debt #16)
docs\DEFERRED_HARDENING_PLAN.md:96:### Approach (minimal first slice ? single self-hosted box)
docs\NEXT_STEPS.md:46:- ? No owned-model implementation until owner greenlights a milestone.
Committed next PLATFORM slice = async job model (single-tenant).
docs\PLATFORM_ARCHITECTURE.md:41:> first infra slice (an async job model) is the on-hold #16 in
docs\PLATFORM_ARCHITECTURE.md:55:| API | `/api/process`, `/api/query`, `/api/compile`,
`/api/config`; no auth | needs auth, per-tenant scoping, async job + history endpoints |
docs\PLATFORM_ARCHITECTURE.md:215:## 5. Orchestration / async jobs
docs\PLATFORM_ARCHITECTURE.md:218:[`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md)
#16** (async job model) ? build
docs\PLATFORM_ARCHITECTURE.md:462:| **P0** | **Async job model, single-tenant** (DEFERRED #16) |
`jobs` table via `user_version` ladder; worker; 202+status endpoints |
docs\PLATFORM_ARCHITECTURE.md:567:- [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) ?
#16 async job model (the P0 infra slice), #7 packaging.
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:165:next platform slice remains the async job model
(single-tenant); only the M0.1?M0.4 foundation is safe to start.*
docs\README.md:10:- [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) ? Design (pending
approval) for the larger deferred items: packaging, train/eval guardrails, and the async job
model (#16, the first infra slice toward the platform).
```

```
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:155:scaffolding (async jobs,
storage/compute backends) is rock-solid AND the golden corpus is healthy. Paid Gemini
docs\archives\checkpoints\2026-06-strategy-foundation\README.md:18:-
**DEFERRED_HARDENING_PLAN.md** ? packaging, train/eval guardrails, async job model (#16) ?
*pending approval*.
docs\archives\checkpoints\2026-06-strategy-foundation\README.md:30:- First implementation slice
= **async job model (DEFERRED #16)**, single-tenant ? **pending explicit go-ahead** (no
implementation started without approval).
docs\archives\checkpoints\2026-06-strategy-foundation\README.md:33:When the first implementation
slice or a quality milestone completes, add a new dated
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:39:Value Layer (exports
EDL/JSON/CSV timelines + basic stats as minimal first slice). ? (feat/value-layer-minimal-
exports + PR #51, merged)
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:40:9. (Stretch) First
slice of async job model.
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:168:- Async job model
slice.
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md:189:  - 3. Value Layer
(minimal first slice: EDL/JSON/CSV timeline exports + basic stats over play segments) ? #51
```

12. user (2026-06-10T06:11:10.299Z)

```
Structured output provided successfully
```

13. assistant (2026-06-10T06:11:16.069Z)

```
Verification complete. The finding was refuted: the committed platform slice (async job model)
is stated consistently across all authority docs, the divergent "packaging-first" signal is a
labeled pre-decision recommendation, and the claimed missing relationship between the async
slice and Phase-0 M0.x is explicitly stated in
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md:164-165`.
```